



CYBERSTER

Internship Program - Batch 2

Week 6 - Mid-Term Assignment

TechNova Solutions: Enterprise Network Design Report

VLSM Subnetting • VLANs • ROAS • Static Routing • ACL Security

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Task:	TechNova Solutions: Mid-Term Network Architecture
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1. Company Scenario & Assignment Requirements

TechNova Solutions requires a fully functional enterprise-grade local area network for a small branch office. The network must accommodate four internal departments with clearly defined host requirements, support remote branch connectivity, and enforce specific inter-departmental reachability policies through Access Control Lists.

Company Scenario: TechNova Solutions

TechNova Solutions is deploying a new local network infrastructure for a small enterprise branch. As the mid-term project for the internship, the task requires designing, subnetting, and configuring the network. A base network block of 192.168.10.0/24 was provided to accommodate four departments.

Base Network: 192.168.10.0/24 • 4 Departments • VLSM Required

#	Task	Deliverable
1	Subnetting & IP Planning (VLSM)	Calculate Network Address, Subnet Mask, First/Last Usable IP, and Broadcast for each department using VLSM from 192.168.10.0/24.
2	Network Architecture & Tier Identification	Design the topology in Cisco Packet Tracer. Label and group devices into Core, Distribution, and Access tiers.
3	VLAN Creation & Switch Configuration	Create VLANs 10/20/30/40 for each department. Assign access ports. Configure trunk uplinks.
4	Static Routing Setup	Deploy a Remote Branch router (192.168.20.0/24) connected via a /30 WAN link (10.0.0.0/30). Configure static routes.
5	Access & Reachability Control	IT: full reachability everywhere. Finance & HR: can talk to each other but blocked from Sales. Verify with ping tests.

2. Subnetting & IP Planning (VLSM)

2.1 VLSM Methodology

Variable Length Subnet Masking (VLSM) allows a single classful network to be divided into subnets of varying sizes, each sized to exactly meet the host requirements of a specific department. This eliminates the waste inherent in classful subnetting. The golden rule of VLSM is to always allocate from largest requirement to smallest, ensuring each subnet begins at the next available address after the previous subnet's broadcast.

Dept	Hosts Needed	VLSM Decision
Sales Dept	60	Needs at least 62 usable hosts → Next power of 2 above 60 is 64 → Block of 64 → /26 (255.255.255.192)
IT Dept	30	Needs exactly 30 usable hosts → Block of 32 → /27 (255.255.255.224)
Finance Dept	14	Needs 14 usable hosts → Block of 16 → /28 (255.255.255.240)
HR Dept	6	Needs 6 usable hosts → Block of 8 → /29 (255.255.255.248)

2.2 Subnet Calculation Results

The VLSM calculations were performed and verified using a CIDR/VLSM calculator. The following three screenshots document the calculation inputs and the resulting subnet allocation table across the 192.168.10.0/24 address space.

CIDR/VLSM Calculator

This VLSM calculator helps you split a network into several smaller subnets.

Step 1:

What is the starting network you would like to split into smaller subnets?
Write the network prefix in slash notation (eg 192.168.1.0/24)

192.168.10.0/24

Step 2:

How many subnets do you need?

Enter a value then click 'Change' 4 Change

Figure 1: VLSM Calculator - Step 1: Base network 192.168.10.0/24, 4 subnets required.

Step 3:

In the left column, enter a name for each subnet (eg 'Sydney Sales')

In the right column, enter how many hosts will be in each subnet

Subnet Names:

Number of hosts:

Sales Department	60
IT Department	30
Finance Department	14
HR Department	6

Figure 2: VLSM Calculator - Step 3: Host count entry for Sales (60), IT (30), Finance (14), HR (6).

The network 192.168.10.0/24 has 254 hosts.
Your subnets need 110 hosts.

Name	Hosts Needed	Hosts Available	Unused Hosts	Network Address	Slash	Mask	Usable Range	Broadcast	Wildcard
Sales Department	60	62	2	192.168.10.0	/26	255.255.255.192	192.168.10.1 - 192.168.10.62	192.168.10.63	0.0.0.63
IT Department	30	30	0	192.168.10.64	/27	255.255.255.224	192.168.10.65 - 192.168.10.94	192.168.10.95	0.0.0.31
Finance Department	14	14	0	192.168.10.96	/28	255.255.255.240	192.168.10.97 - 192.168.10.110	192.168.10.111	0.0.0.15
HR Department	6	6	0	192.168.10.112	/29	255.255.255.248	192.168.10.113 - 192.168.10.118	192.168.10.119	0.0.0.7

Figure 3: VLSM Calculator - Results: Complete subnet allocation table with network addresses, masks, usable ranges, broadcasts, and wildcard masks.

VLSM Subnet Summary Table - 192.168.10.0/24					
Department	VLAN	Network Address	Subnet Mask	Usable Range	Broadcast
Sales	10	192.168.10.0	/26 - 255.255.255.192	192.168.10.1 - .62	192.168.10.63
IT	20	192.168.10.64	/27 - 255.255.255.224	192.168.10.65 - .94	192.168.10.95
Finance	30	192.168.10.96	/28 - 255.255.255.240	192.168.10.97 - .110	192.168.10.111
HR	40	192.168.10.112	/29 - 255.255.255.248	192.168.10.113 - .118	192.168.10.119

Note: The First Usable IP of each subnet was assigned to the Main Router sub-interfaces to serve as the default gateway for that VLAN (e.g., 192.168.10.1 for Sales, 192.168.10.65 for IT, 192.168.10.97 for Finance, 192.168.10.113 for HR).

3. Network Architecture & Tier Identification

The physical and logical topology was designed using the Cisco Three-Tier Hierarchical Model to isolate fault domains, streamline traffic flow, and enable independent scaling of each tier. Devices were carefully placed based on their Layer 2 or Layer 3 capabilities and their functional role within the enterprise.

3.1 Initial Topology Design

The initial topology layout was constructed in Cisco Packet Tracer, establishing the physical connectivity between all devices before configuration. This baseline confirms all cable connections are correct before any logical configuration is applied.

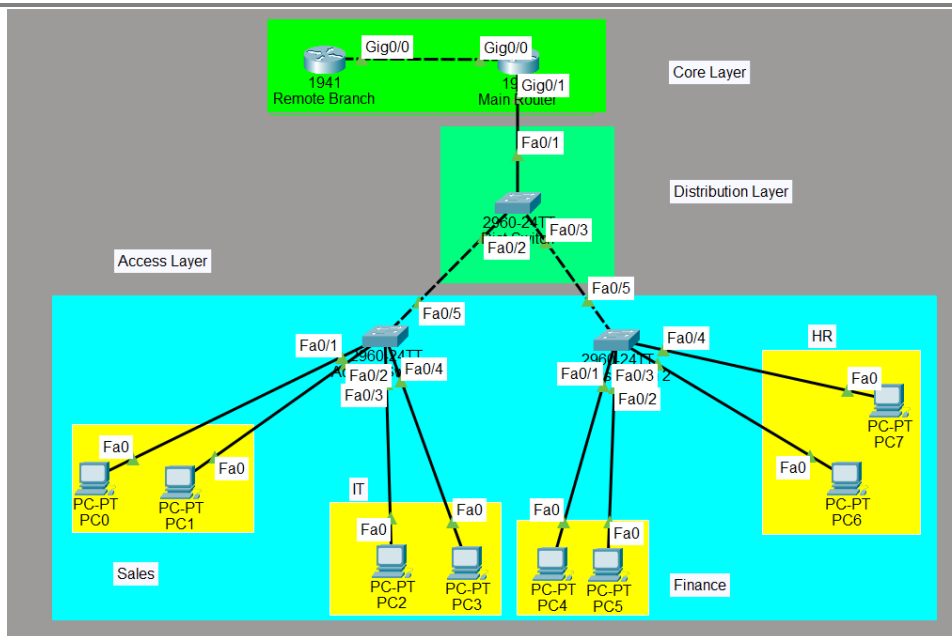


Figure 4: Initial Cisco Packet Tracer topology - physical cabling complete, colour-coded layer groupings visible.

3.2 Final Labelled Topology

The final topology with all tier labels applied is shown below. The Core, Distribution, and Access layers are clearly delineated with background colour groupings, matching the hierarchical model requirement of the assignment.

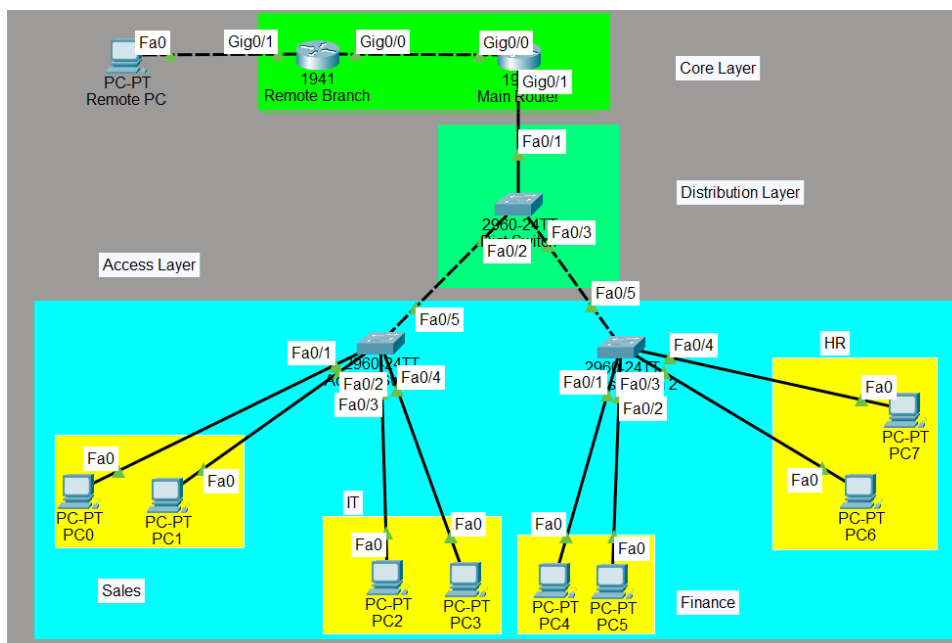


Figure 5: Final labelled topology - Core (green), Distribution (green), Access (cyan) with departmental VLAN colour groupings (Sales, IT, Finance, HR in yellow).

3.3 Device Role Justification

Tier	Device(s)	Layer	Why Placed Here
Core / Aggregation Block	Main Router (1941) Remote Branch Router (1941)	Layer 3 - Routing + inter-VLAN gateway	Only the router has the Layer 3 intelligence to perform inter-VLAN routing (ROAS), apply Extended ACLs, and exchange static routes with the Remote Branch.
Distribution Block	Dist Switch (2960-24TT)	Layer 2 - Trunk aggregation	Aggregates trunk uplinks from both Access Switches and passes all VLAN-tagged frames to the router. Does not connect directly to any end-user device.
Access Block	Access Switch 1 (2960) Access Switch 2 (2960) All PCs (PC0-PC7) Remote PC	Layer 2 - Edge access	Provides port-level endpoint connectivity. VLAN membership is enforced here via access ports. Trunk ports carry tagged frames up to the Distribution Switch.

4. Switch Configurations - VLANs & Trunking

To achieve logical segmentation of the four departments, VLANs were created and applied across the switching infrastructure. Each access switch is responsible for its own set of VLANs, with a dedicated trunk port connecting upward to the Distribution Switch.

4.1 Access Switch 1 - Sales (VLAN 10) & IT (VLAN 20)

Parameter	VLAN 10 - Sales	VLAN 20 - IT
VLAN ID	10	20
VLAN Name	Sales	IT
Access Ports	Fa0/1 – Fa0/2	Fa0/3 – Fa0/4
Trunk Port	Fa0/5 (uplink to Dist Switch)	Fa0/5 (shared trunk)
Connected Hosts	PC0, PC1	PC2, PC3

Access Switch 1

Physical Config CLI Attributes

```

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Create VLANs
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#! Assign Sales Ports
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#! Assign IT Ports
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#! Configure Trunk Port
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk

Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#

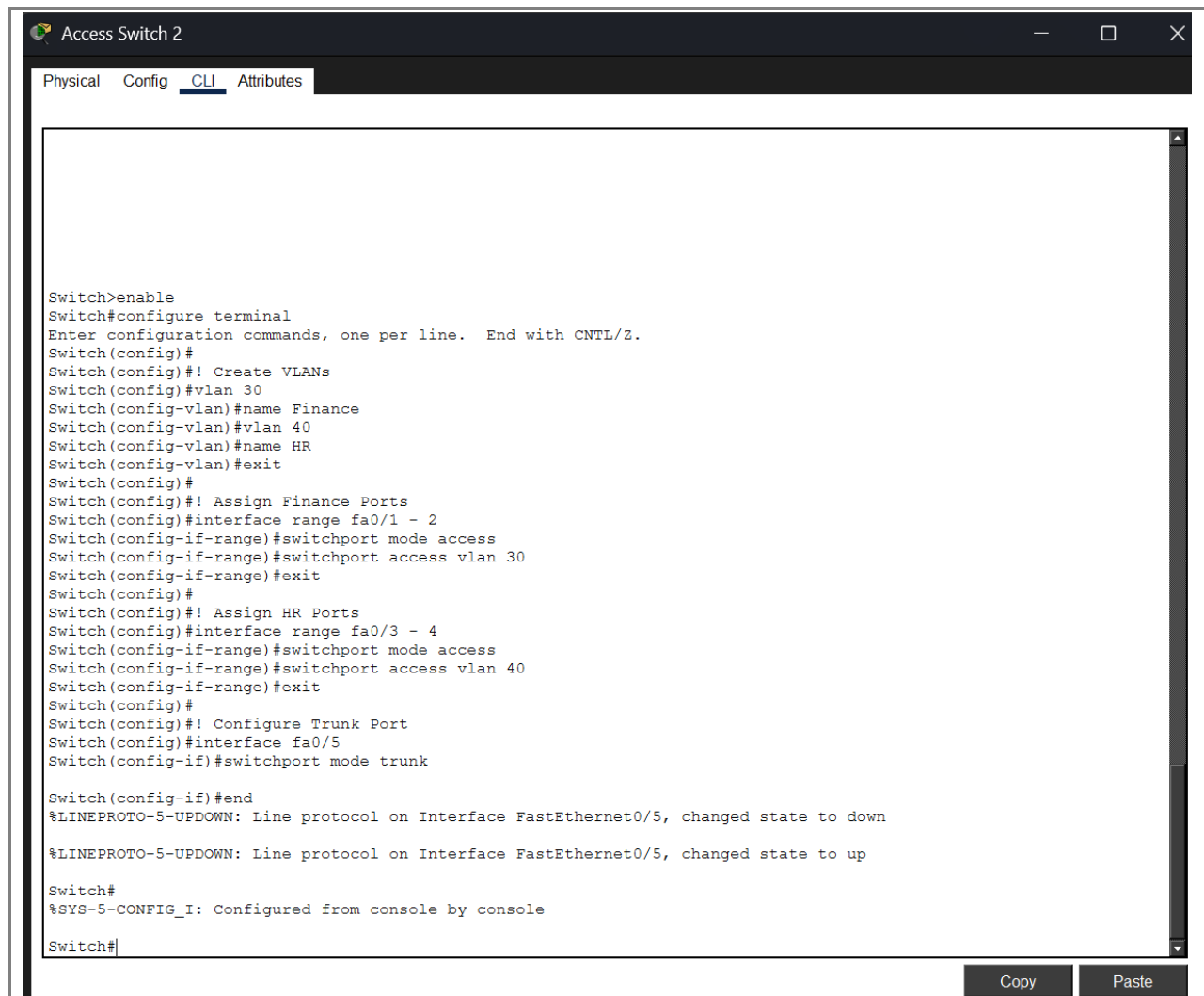
```

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Figure 6: Access Switch 1 CLI - VLAN 10 (Sales) on Fa0/1–2, VLAN 20 (IT) on Fa0/3–4, trunk on Fa0/5.

4.2 Access Switch 2 - Finance (VLAN 30) & HR (VLAN 40)

Parameter	VLAN 30 — Finance	VLAN 40 — HR
VLAN ID	30	40
VLAN Name	Finance	HR
Access Ports	Fa0/1 – Fa0/2	Fa0/3 – Fa0/4
Trunk Port	Fa0/5 (uplink to Dist Switch)	Fa0/5 (shared trunk)
Connected Hosts	PC4, PC5	PC6, PC7



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Create VLANs
Switch(config)#vlan 30
Switch(config-vlan)#name Finance
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#! Assign Finance Ports
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#! Assign HR Ports
Switch(config)#interface range fa0/3 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#! Configure Trunk Port
Switch(config)#interface fa0/5
Switch(config-if)#switchport mode trunk

Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

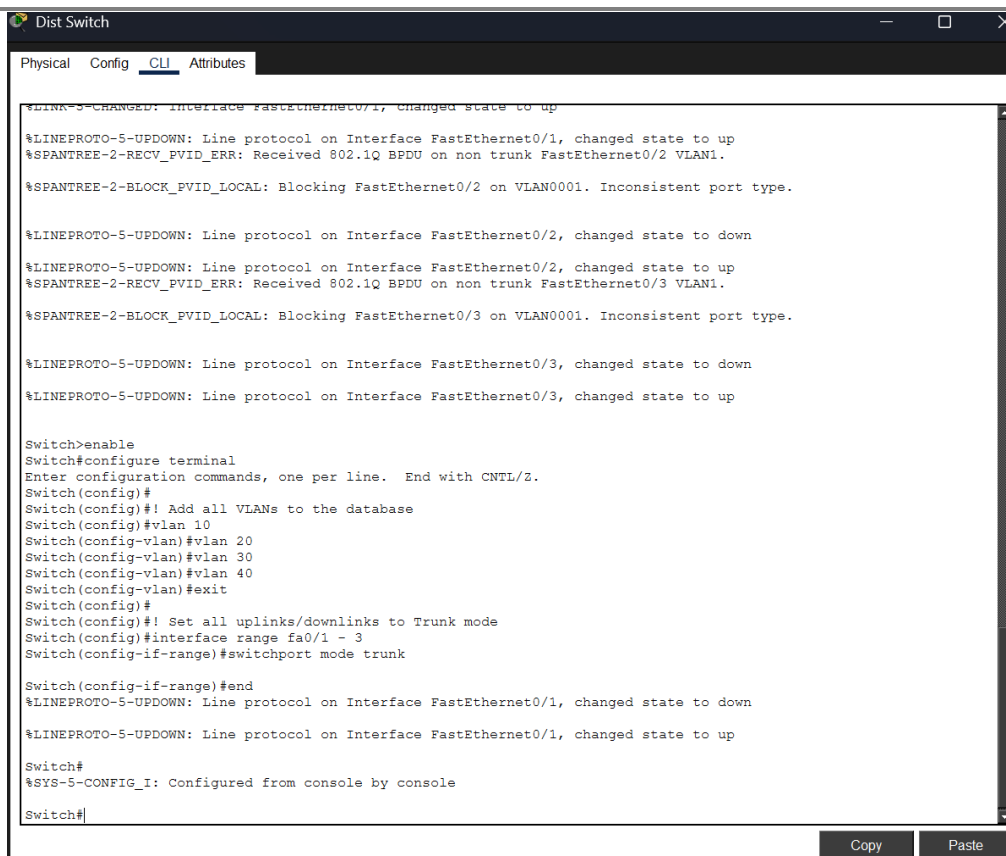
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Figure 7: Access Switch 2 CLI - VLAN 30 (Finance) on Fa0/1–2, VLAN 40 (HR) on Fa0/3–4, trunk on Fa0/5.

4.3 Distribution Switch

The Distribution Switch does not connect to any end-user devices directly. Its sole function is to aggregate trunk links from both Access Switches and pass all VLAN-tagged traffic to the Main Router via a trunk uplink. All four VLANs (10, 20, 30, 40) are added to its local VLAN database, and all active interfaces (Fa0/1, Fa0/2, Fa0/3) are set to trunk mode.



```
Dist Switch
Physical Config CLI Attributes

%LINK-3-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/3 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/3 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Add all VLANs to the database
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#vlan 30
Switch(config-vlan)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#! Set all uplinks/downlinks to Trunk mode
Switch(config)#interface range fa0/1 - 3
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

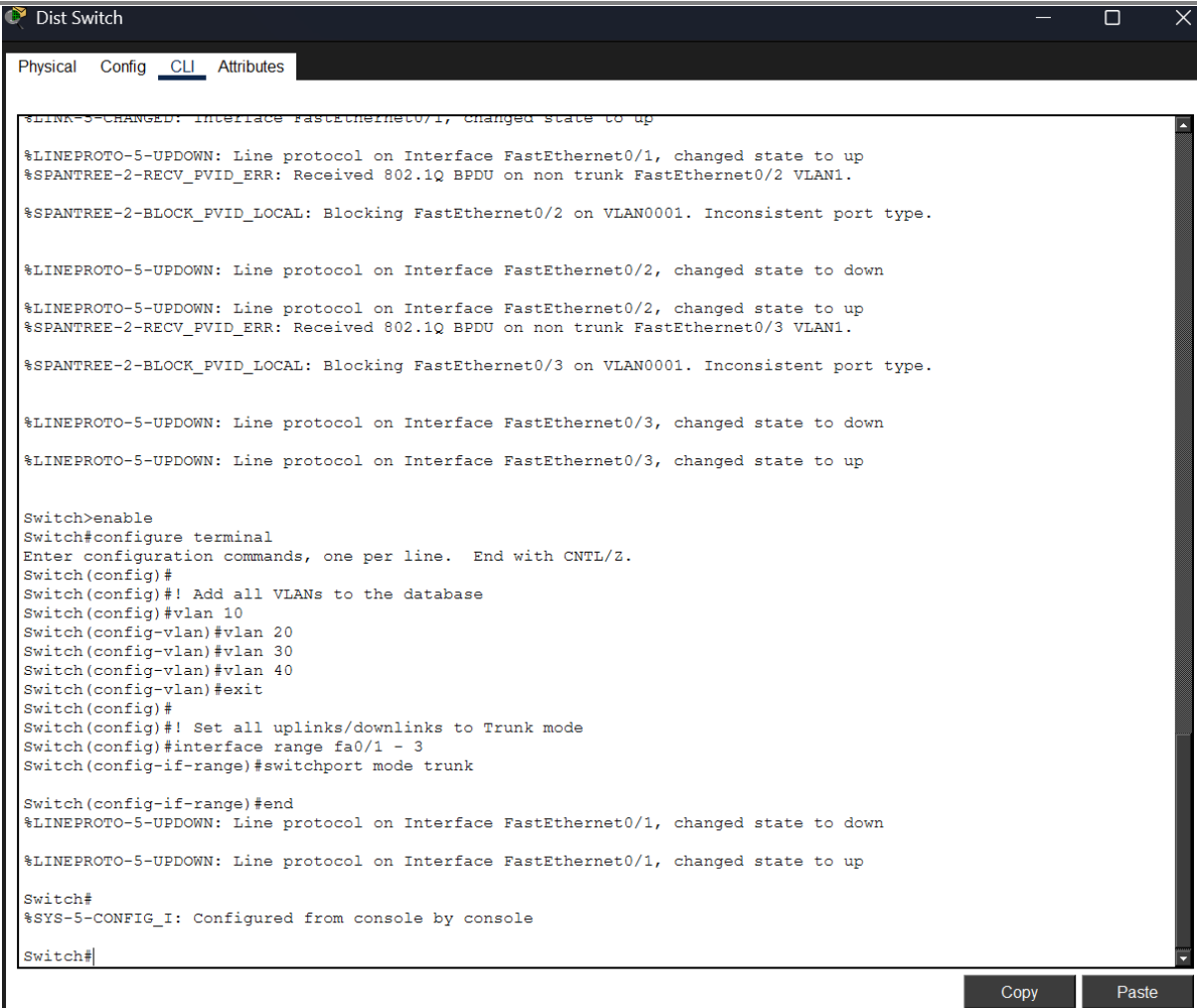
Figure 8: Distribution Switch CLI - VLANs 10/20/30/40 created; Fa0/1–3 set to trunk mode (PVID errors resolved after trunk applied).

5. Router Configurations - ROAS & Remote Branch

5.1 Main Router - ROAS Sub-Interfaces

To enable inter-VLAN communication without a Layer 3 switch, the Router-on-a-Stick (ROAS) technique was applied on the Main Router's Gig0/1 interface. Four logical sub-interfaces were created, each encapsulated with an 802.1Q tag matching its VLAN ID, and each assigned the First Usable IP of its respective subnet as the default gateway.

Sub-Interface	VLAN	Encapsulation / IP	Subnet Mask	Gateway For
Gig0/1.10	VLAN 10	dot1Q 10 / 192.168.10.1	/26	Sales
Gig0/1.20	VLAN 20	dot1Q 20 / 192.168.10.65	/27	IT
Gig0/1.30	VLAN 30	dot1Q 30 / 192.168.10.97	/28	Finance
Gig0/1.40	VLAN 40	dot1Q 40 / 192.168.10.113	/29	HR



```
Dist Switch
Physical Config CLI Attributes

%LINK-3-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%SPANTRIE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/2 VLAN1.
%SPANTRIE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/2 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%SPANTRIE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/3 VLAN1.
%SPANTRIE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/3 on VLAN0001. Inconsistent port type.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Add all VLANs to the database
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#vlan 30
Switch(config-vlan)#vlan 40
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#! Set all uplinks/downlinks to Trunk mode
Switch(config)#interface range fa0/1 - 3
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

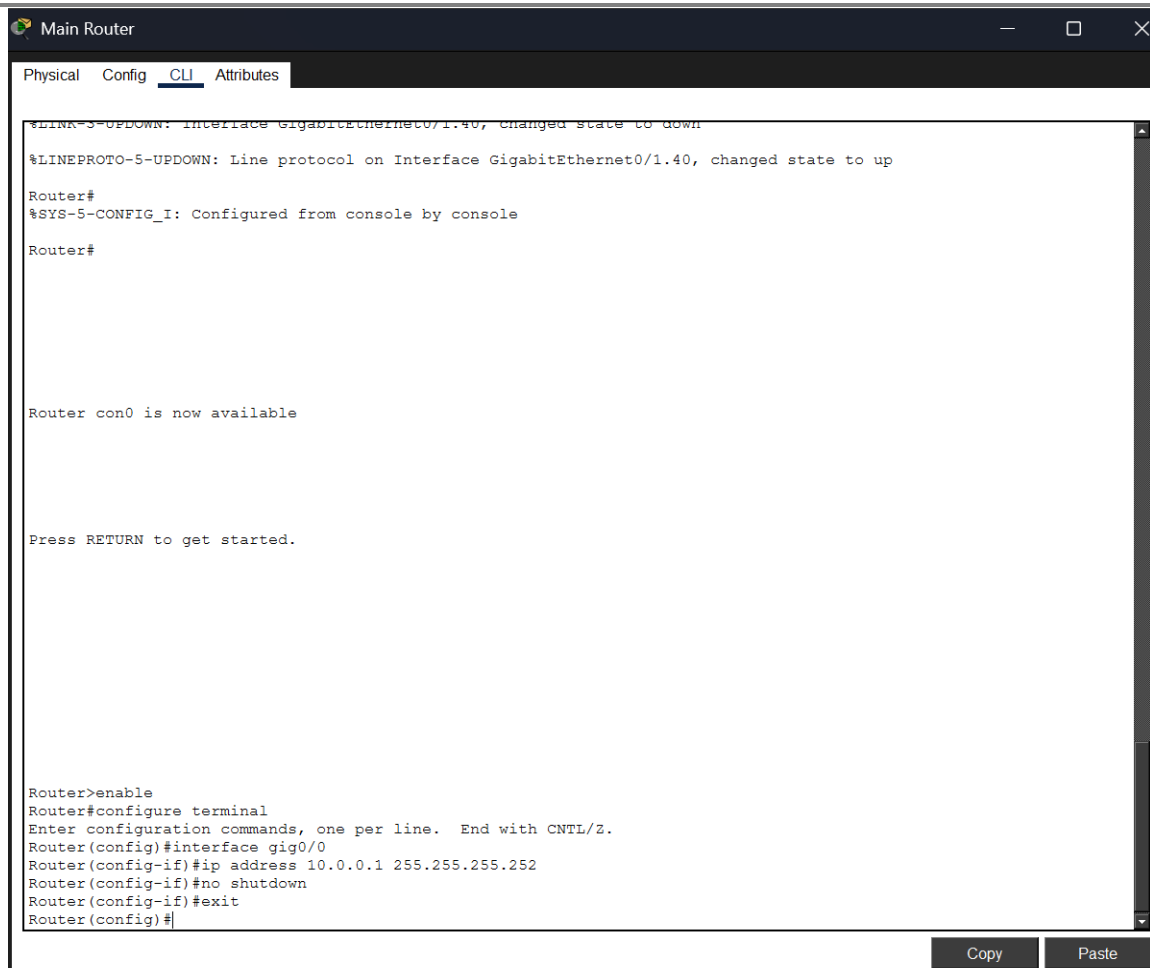
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Figure 9: Main Router CLI - ROAS sub-interfaces Gig0/1.10/20/30/40 configured with 802.1Q encapsulation and department gateway IPs. Sub-interfaces confirmed up.

5.2 Main Router - Remote Branch WAN Link

A dedicated WAN link was established between the Main Router and the Remote Branch router using a /30 point-to-point subnet (10.0.0.0/30). The Main Router's Gig0/0 interface was assigned 10.0.0.1/30, and the Remote Branch's Gig0/0 was assigned 10.0.0.2/30. This creates a minimal, non-wasteful 2-host subnet for the router-to-router link.



```

Main Router
Physical Config CLI Attributes

%LINK-3-UPDOWN: Interface GigabitEthernet0/1.40, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#

Router con0 is now available

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#

```

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Figure 10: Main Router CLI - Gig0/0 configured with WAN address 10.0.0.1/30 for the Remote Branch point-to-point link.

5.3 Remote Branch Router

The Remote Branch router simulates an external office location. Its Gig0/0 interface (10.0.0.2/30) faces the Main Router over the WAN link, while its Gig0/1 interface (192.168.20.1/24) acts as the default gateway for the Remote LAN (192.168.20.0/24), which hosts the Remote PC (192.168.20.2).

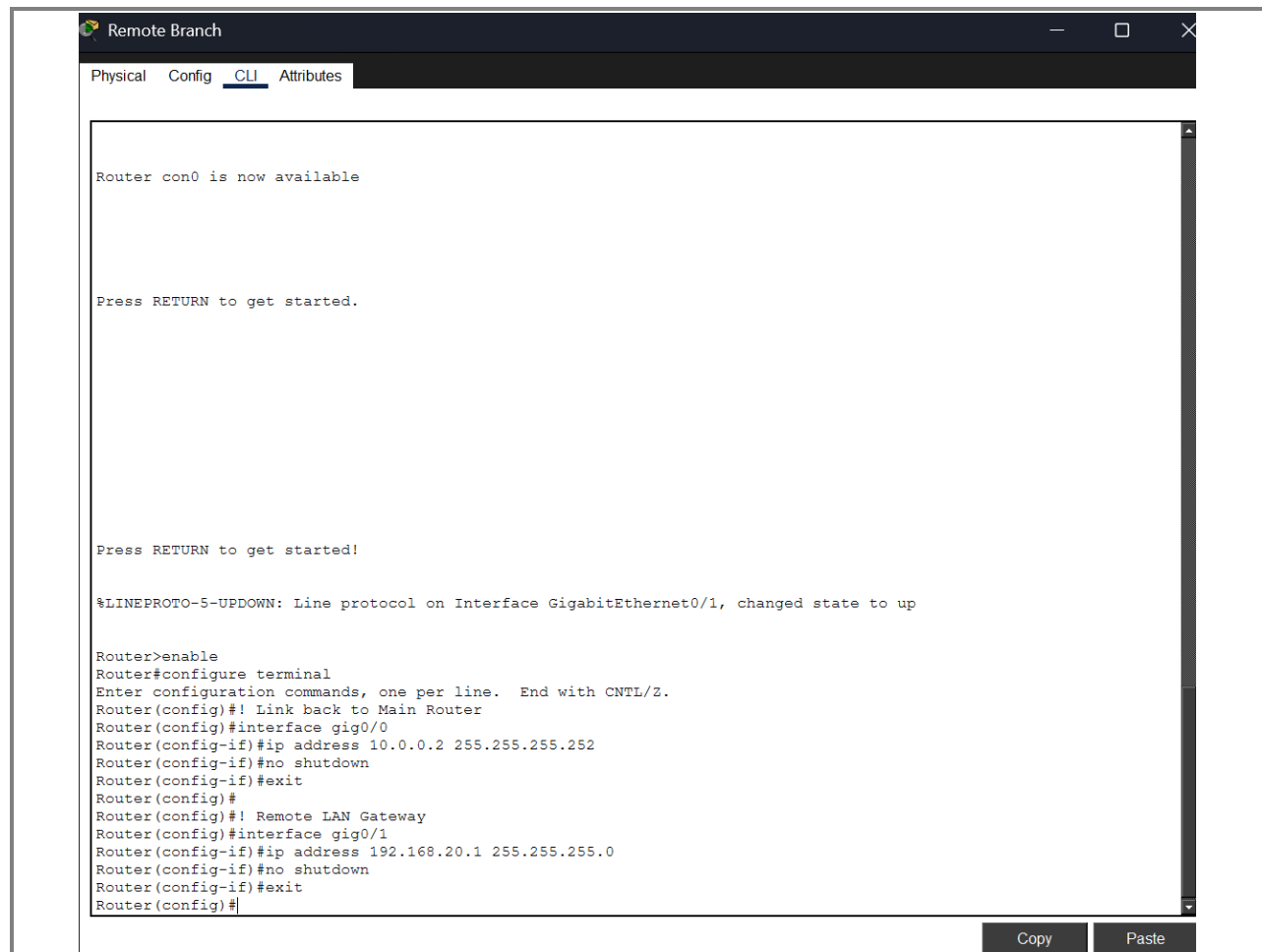


Figure 11: Remote Branch Router CLI - Gig0/0 (10.0.0.2) WAN interface; Gig0/1 (192.168.20.1) local LAN gateway for the remote subnet.

6. Reachability & Access Control - Static Routing + ACLs

To satisfy the specific security and reachability policy of TechNova Solutions, two mechanisms were combined: Static Routing for inter-site reachability, and an Extended Access Control List (ACL) to enforce departmental traffic restrictions.

Policy	Implementation
IT → Full Reachability	No ACL restrictions applied to VLAN 20 traffic. IT can reach all VLANs and the Remote Branch freely via the ROAS router.
Finance → HR (Allowed)	Finance (192.168.10.96/28) and HR (192.168.10.112/29) can reach each other - ACL only blocks their access to Sales.
Finance → Sales (BLOCKED)	ACL 100 explicitly denies traffic from 192.168.10.96/28 (Finance wildcard 0.0.0.15) to the Sales subnet (192.168.10.0/26).

HR → Sales (BLOCKED)	ACL 100 explicitly denies traffic from 192.168.10.112/29 (HR wildcard 0.0.0.7) to the Sales subnet.
All Others (Permitted)	ACL 100 ends with 'permit ip any any' - all other traffic not matched by the deny statements is forwarded normally.
ACL Application Point	ACL 100 is applied OUTBOUND on Gig0/1.10 (the Sales sub-interface) - traffic is checked only when exiting toward Sales.

6.1 Remote Branch Default Route

A static default route (ip route 0.0.0.0 0.0.0.0 10.0.0.1) was configured on the Remote Branch router. This instructs the Remote Branch to forward all traffic destined for unknown networks (including any internal enterprise subnet) toward the Main Router at 10.0.0.1.

```

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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Compiled Thurs 5-Jan-12 15:41 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

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compliance with U.S. and local country laws. By using this product you
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If you require further assistance please contact us by sending email to
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Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

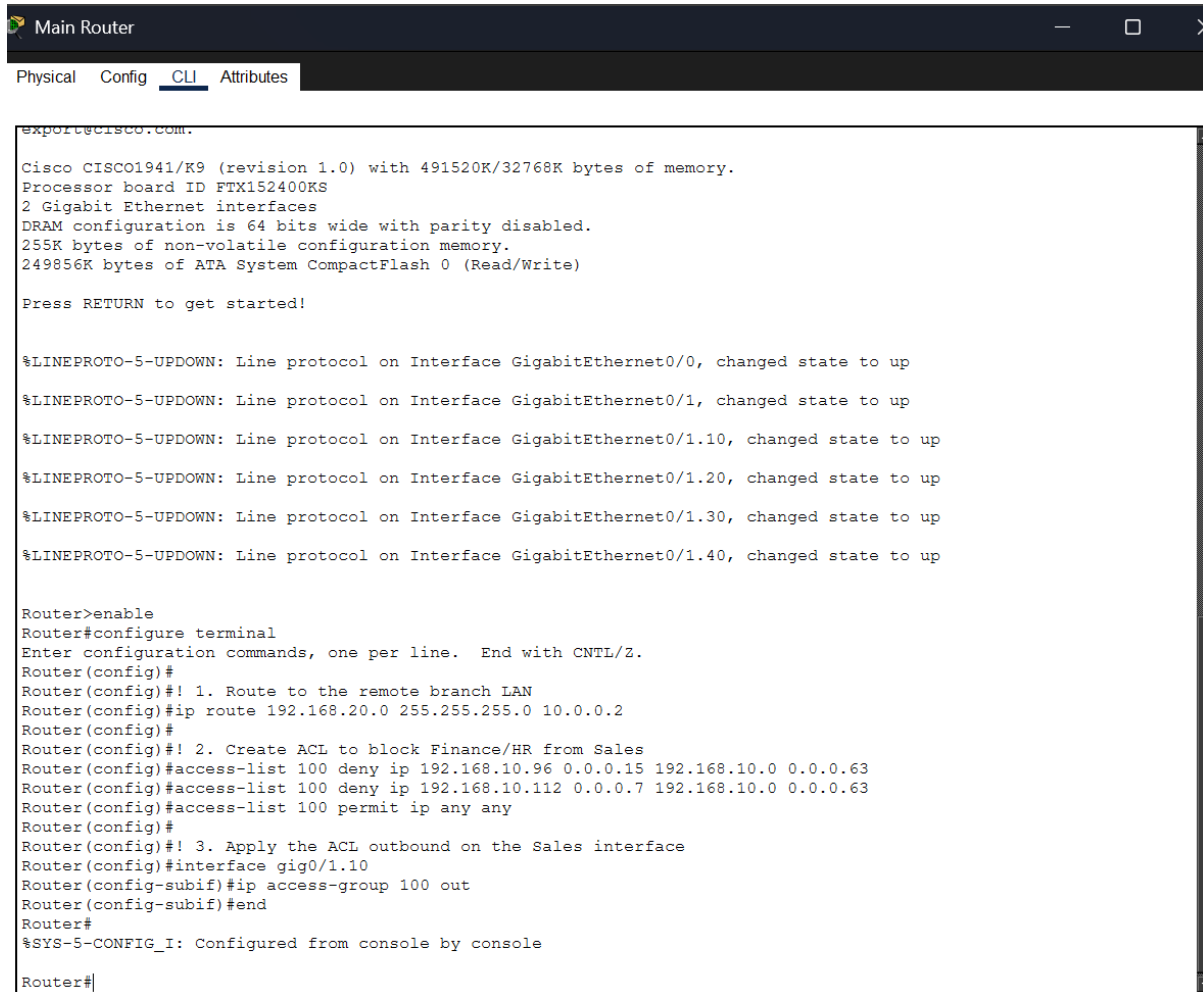
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure 12: Remote Branch Router - default static route 0.0.0.0/0 via 10.0.0.1 ensures all outbound traffic is forwarded to the Main Router.

6.2 Main Router - Static Route + Extended ACL

On the Main Router, a static route was added to reach the Remote Branch LAN (192.168.20.0/24) via the next-hop 10.0.0.2. Additionally, Extended ACL 100 was created with three statements: deny Finance to Sales, deny HR to Sales, and permit all other traffic. The ACL was then applied outbound on the Sales sub-interface (Gig0/1.10).



```
export@cisco.com:
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! 1. Route to the remote branch LAN
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#
Router(config)#! 2. Create ACL to block Finance/HR from Sales
Router(config)#access-list 100 deny ip 192.168.10.96 0.0.0.15 192.168.10.0 0.0.0.63
Router(config)#access-list 100 deny ip 192.168.10.112 0.0.0.7 192.168.10.0 0.0.0.63
Router(config)#access-list 100 permit ip any any
Router(config)#
Router(config)#! 3. Apply the ACL outbound on the Sales interface
Router(config)#interface gig0/1.10
Router(config-subif)#ip access-group 100 out
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Figure 13: Main Router CLI - static route to 192.168.20.0/24 via 10.0.0.2; ACL 100 with deny Finance/HR to Sales, permit any; ACL applied outbound on Gig0/1.10.

7. Connectivity Verification - Ping Tests

All ping tests were conducted from representative PCs within each department to confirm that the routing, VLAN, and ACL configurations are operating exactly as designed. The first packet loss in each test is normal - it is consumed by the ARP resolution process.

7.1 IT Department - Full Reachability

The IT Department (VLAN 20, subnet 192.168.10.64/27) must have unrestricted access to all other departments and the Remote Branch. Tests were initiated from an IT PC (192.168.10.66).

Source	Destination	Expected	Result
IT PC (192.168.10.66)	Sales PC - 192.168.10.2	SUCCESS (IT not blocked)	✓ 3/4 packets - TTL=127
IT PC (192.168.10.66)	Remote PC - 192.168.20.2	SUCCESS (full reachability)	✓ 3/4 packets - TTL=127

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.2: bytes=32 time=2ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Figure 14a: IT PC → Sales (192.168.10.2) - 3/4 success (first packet = ARP miss). TTL=127 confirms routing via Main Router.

```

PC2
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126
Reply from 192.168.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

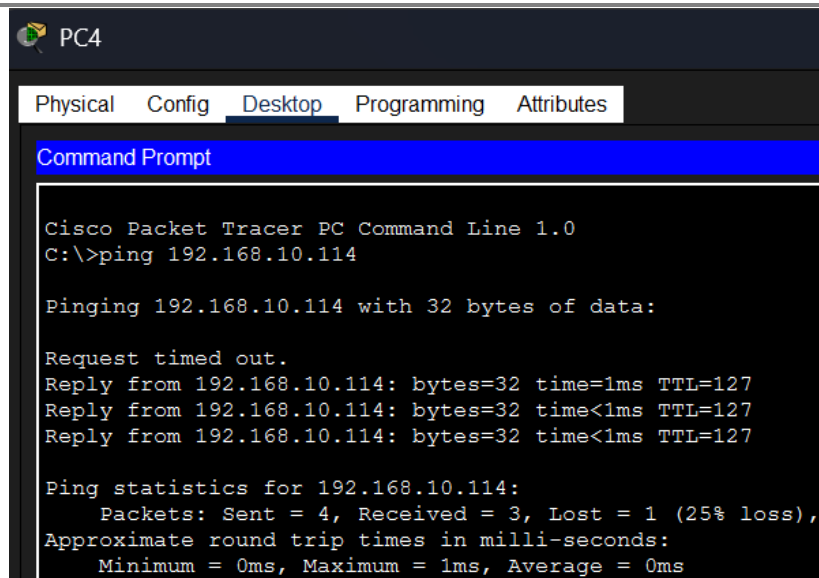
```

Figure 14b: IT PC → Remote PC (192.168.20.2) - 3/4 success. TTL=127 confirms two routing hops (Main Router + Remote Branch).

7.2 Finance & HR - Restricted Reachability

The Finance (VLAN 30 - 192.168.10.96/28) and HR (VLAN 40 - 192.168.10.112/29) departments must be able to communicate with each other freely, but all traffic from Finance or HR destined for the Sales subnet must be blocked by ACL 100 on the Main Router.

Source	Destination	Expected	Result
Finance PC (192.168.10.98)	HR PC - 192.168.10.114	SUCCESS (Finance→HR allowed)	✓ 2/4 packets - TTL=126
Finance PC (192.168.10.98)	Sales PC - 192.168.10.2	BLOCKED by ACL 100	X Dest. Host Unreachable - 100% loss



PC4

Physical Config Desktop Programming Attributes

Command Prompt

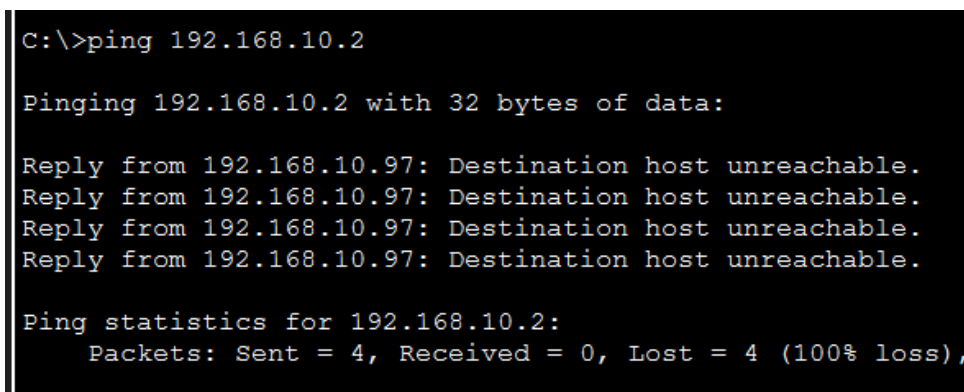
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.114

Pinging 192.168.10.114 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.114: bytes=32 time=1ms TTL=127
Reply from 192.168.10.114: bytes=32 time<1ms TTL=127
Reply from 192.168.10.114: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.114:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 15a: Finance PC → HR (192.168.10.114) - 2/4 success (first 2 pkts = ARP+route learning).
Communication between Finance and HR is allowed.



```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 15b: Finance PC → Sales (192.168.10.2) - 100% blocked. Main Router returns 'Destination host unreachable' - ACL 100 is enforcing the policy correctly.

How ACL 100 Works

```
access-list 100 deny ip 192.168.10.96 0.0.0.15 192.168.10.0 0.0.0.63
→ Denies: any source from Finance (.96–.111) to Sales (.0–.63)
access-list 100 deny ip 192.168.10.112 0.0.0.7 192.168.10.0 0.0.0.63
→ Denies: any source from HR (.112–.119) to Sales (.0–.63)
access-list 100 permit ip any any
→ Permits: all other traffic (IT, Sales self-traffic, Remote Branch, etc.)
```

8. Conclusion

This mid-term assignment successfully demonstrated the complete design, configuration, and validation of an enterprise-grade small branch network for TechNova Solutions using Cisco Packet Tracer. Five distinct technical competencies were exercised and verified:

✓	Achievement
✓	VLSM Subnetting: The 192.168.10.0/24 block was efficiently divided into four variable-length subnets - /26 (Sales), /27 (IT), /28 (Finance), and /29 (HR) - eliminating IP waste and precisely matching each department's host requirements.
✓	Hierarchical Network Design: The topology was built and correctly labelled across three tiers: Core/Aggregation (Main Router + Remote Branch Router), Distribution (Dist Switch), and Access (Access SW1, SW2, and all PCs). Each device placement was justified by its Layer 2/3 capabilities.
✓	VLAN Segmentation: Four VLANs (10 Sales, 20 IT, 30 Finance, 40 HR) were created and applied across both Access Switches. Trunk ports correctly carry all tagged VLAN frames between the Access layer, Distribution Switch, and Main Router.
✓	Router-on-a-Stick (ROAS): Inter-VLAN routing was achieved on the Main Router using 802.1Q sub-interfaces (Gig0/1.10 through .40), enabling communication between all four departmental VLANs through a single physical router interface.
✓	Static Routing: A point-to-point /30 WAN link (10.0.0.0/30) was established between the Main Router and the Remote Branch. Static routes were configured on both sides to ensure bi-directional reachability between the enterprise LAN and the Remote LAN (192.168.20.0/24).
✓	Extended ACL Security: ACL 100 enforces TechNova's reachability policy - Finance and HR are blocked from reaching Sales (100% packet drop confirmed), while their mutual communication is preserved. IT retains full unrestricted access across all segments. All policies were verified with live ping tests.

This mid-term assignment brought together all networking concepts covered in Weeks 1 through 5 of the CYBERSTER internship - from Layer 1 physical cabling and Layer 2 VLAN segmentation to Layer 3 routing, subnetting mathematics, and security policy enforcement via ACLs. The successful verification of all ping tests confirms that the TechNova Solutions network is fully operational, correctly segmented, and properly secured - ready to serve the enterprise's four departments and its remote branch location.